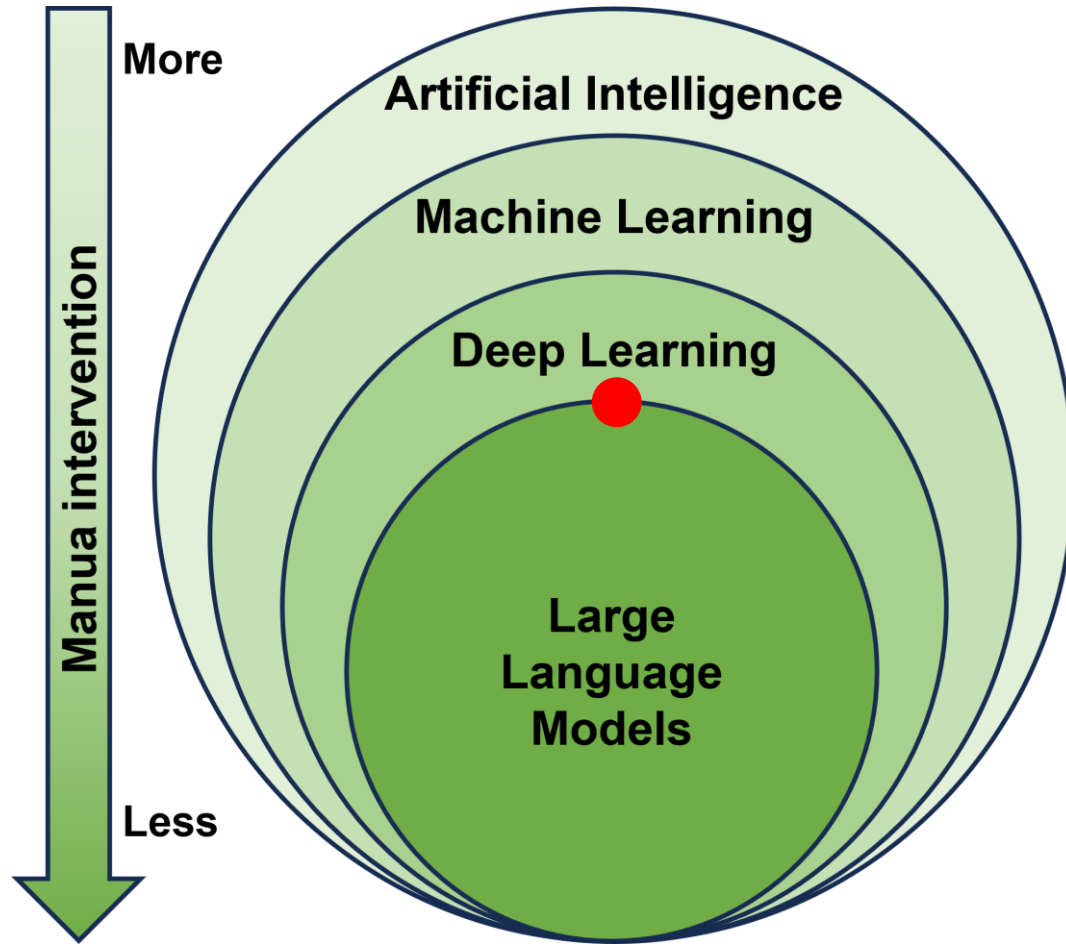


# Introduction to Genomic Large Language Models (LLMs)

**Tutorial VT1: Genomic LLMs in Practice  
– A Hands-On Introduction with Hugging Face**

Presented by Prof. Jean-Christophe (JC) Nebel

# AI-Driven Natural Language Processing for Genomics



The Transformer made LLMs possible by enabling models to scale efficiently and learn rich language representations.



# A Pivotal Shift: The Transformer

Text translation used to rely on RNN models that compress an entire sentence into a **single fixed vector** and **process it sequentially**.

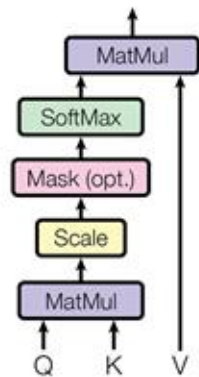
- **Attention mechanism (2014) [9]: Foundation of the Transformer**
  - Problem: With long sentences, important information in the vector can be lost or diluted.
  - Solution: The model dynamically “pays attention to” to the most relevant parts of the input sequence when generating each word.
- **“Attention is all you need” (2017) [10]: The Transformer**
  - Remaining issue: Sequential processing limits efficiency and scalability.
  - Key idea: The model attends to all words simultaneously, enabling parallel processing and more effective learning of relationships.

# Attention Mechanism

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad [10]$$

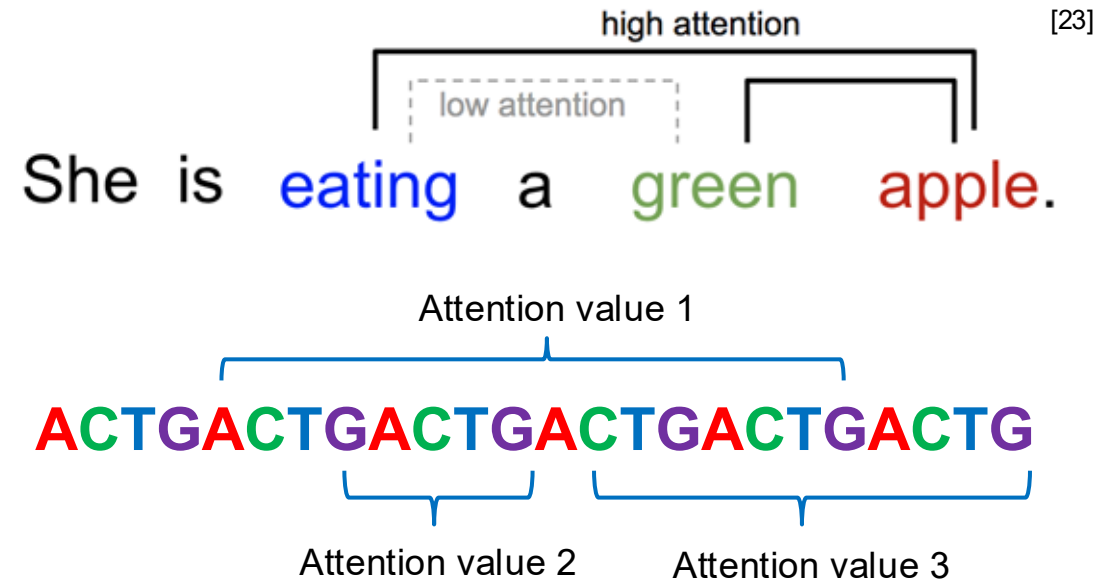
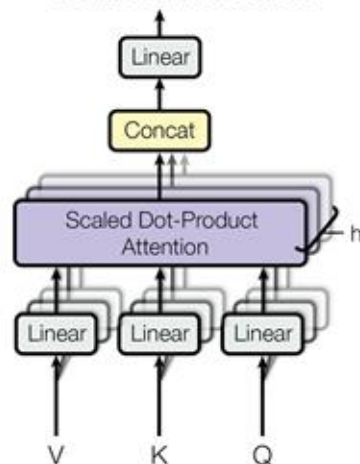
- **Scaled dot-product attention**
  - Most basic form of self-attention, which allows a model to weigh up the importance of different parts of an input sequence
  - Input consists of queries (Q) and keys (K) of dimension  $d_k$ , and values (V) of dimension  $d_v$
- **Multi-head attention**
  - Runs multiple self-attention processes in parallel
  - Allows the model to jointly attend to information from different representation subspaces at different positions

Scaled Dot-Product Attention



[10]

Multi-Head Attention

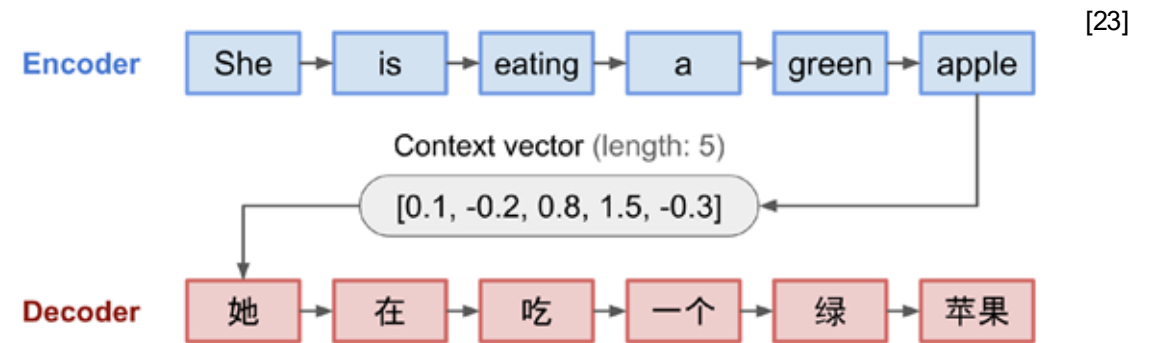
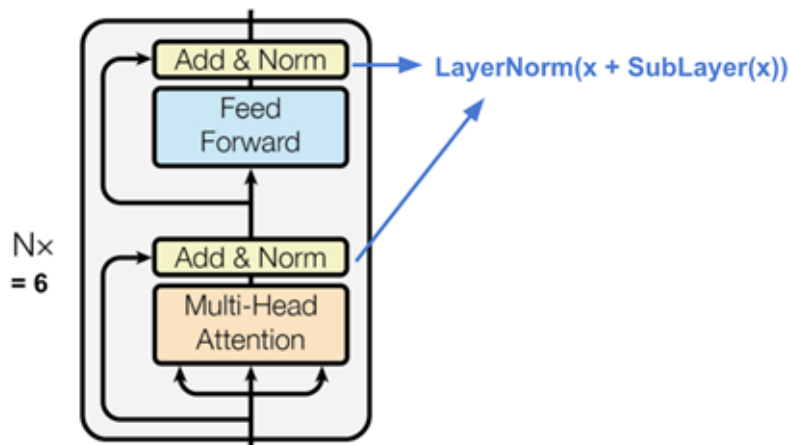


[23]

# Encoder vs Decoder

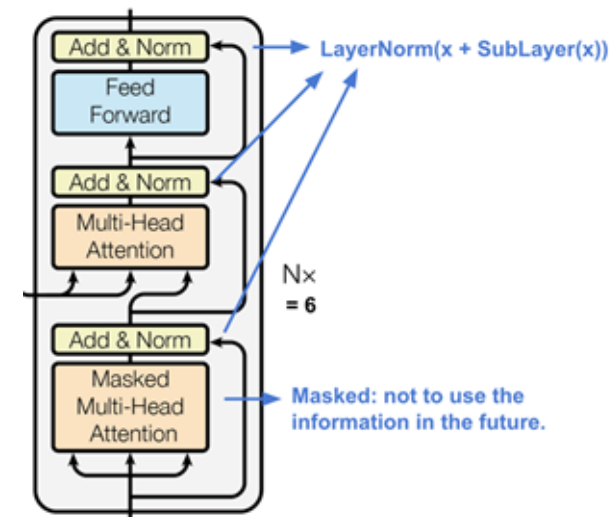
## Encoder

- Stack of identical layers: each containing a multi-head self-attention mechanism and a fully connected feed-forward network
- Generates an attention-based representation of the data
- Can find specific information from long context



## Decoder

- Stack of identical layers: in addition to the sub-layers in the encoder, each of the decoder layers contains a third sub-layer which performs multi-head attention over the output of the encoder stack
- Retrieves information from the encoded representation



# Tokenisers – Introduction

Deep learning reduces manual feature design, but it does not eliminate assumptions. In genomic LLMs, those assumptions are embedded in choices like **tokenisation** and model architecture.

**Tokenisation:** the process of converting raw DNA sequences (ATGCGTACGTTAGC...) into discrete “tokens” that a model can understand and process, e.g.,

- Single-nucleotide tokens (A | T | G | C ...): Simple, no assumption...  
but sequences can be very long, and it is harder to learn higher-order patterns
- Codon-based tokens (ATG | CGT ...): Biologically meaningful in coding regions only
- K-mers (fixed-length chunks, e.g., ATGC | GTAC ...): Split sequence into non-overlapping or overlapping substrings
- Data-driven tokenisation (ATG | CGTA | CGT ...): e.g., **BPE**, WordPiece, & SentencePiece

# Tokenisers – K-mers

The sequence is split into substrings of a fixed length  $k$  (called  $k$ -mers)

**Non-overlapping**  $k$ -mers (used by Nucleotide Transformer,  $k=6$ ):

Pros:

- Reduce sequence length
- Simpler computational processing

Cons:

- May miss motifs spanning token boundaries
- Less robust to sequence variations (indels create dramatic changes)

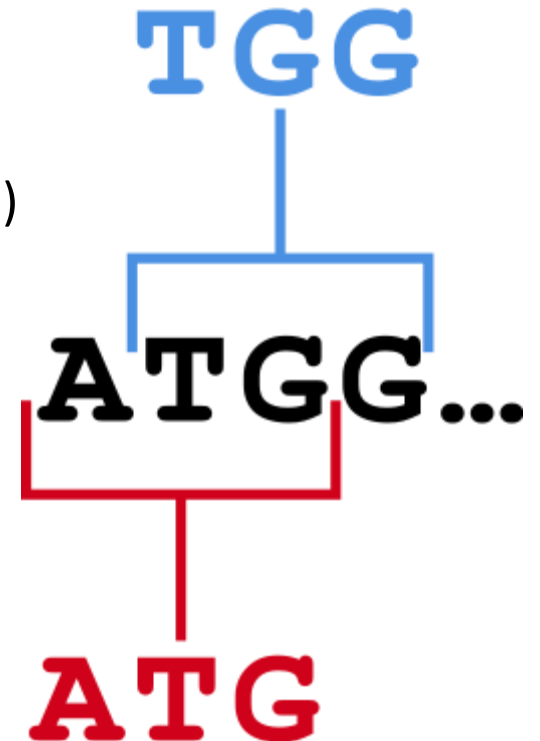
**Overlapping**  $k$ -mers (used by DNABERT-1,  $k = 3$  to  $6$ ):

Pros:

- Captures local motifs effectively
- Provides richer context via overlap

Cons:

- Introduce redundancy and longer sequences
- Higher computational and memory load



# Tokenisers – Byte-Pair Encoding (BPE) (used by DNABERT-2 [20], GROVER [21])

With fixed-length tokens, token frequencies become very heterogeneous

- Genomes are not random strings, e.g., many genomes are A/T-rich because CG dinucleotides are prone to mutation

**BPE** assumes that frequently occurring substrings are meaningful.

It builds progressively larger tokens from frequent sequence content by iteratively merging the most common adjacent token pairs.

Let's encode AAAGTAAAGAC using BPE:

Sequence

AAAGTAAAGAC

XAGTXAGAC

XYTXYAC

ZTZAC

i.e., [AAAG][T][AAAG][A][C]

Tokens

{A, G, T, C}

{A, G, T, C, X}, where X = AA

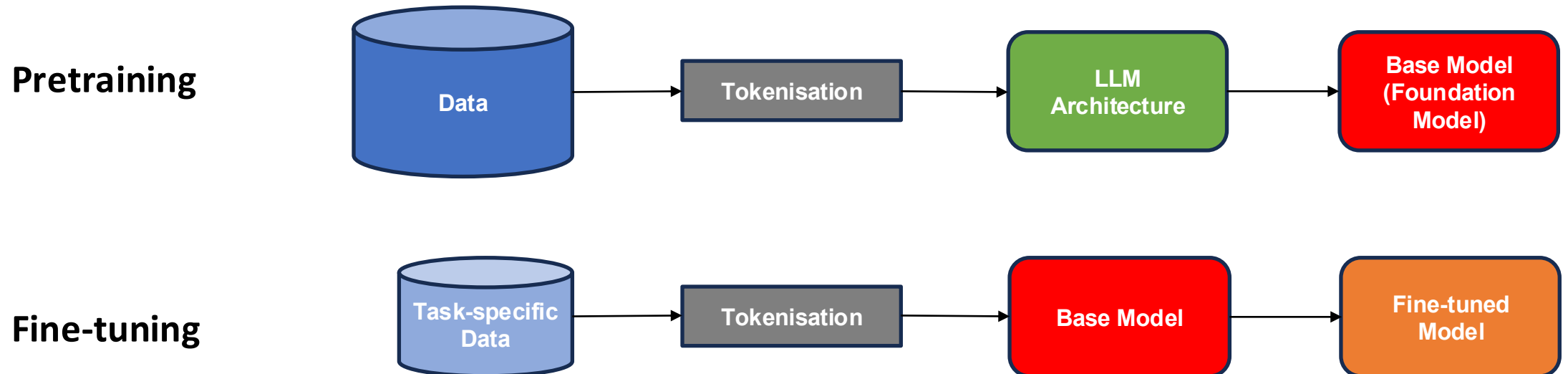
{A, G, T, C, X, Y}, where Y = AG & X = AA

{A, G, T, C, X, Y, Z}, where Z = XY, Y = AG and X = AA

# Large Language Models (from 2018)

Large-scale models built on the Transformer architecture

- Have usually billions of parameters
  - Trained in an unsupervised manner (no need for labels) on large corpora of sequence data requiring significant amount of processing power
- ➔ Ability to accurately model long-range dependencies within sequences



# GPT-1 - Generative Pre-trained Transformer (2018) [12]

‘General language generator’ by OpenAI

- Architecture
  - Decoder-only transformer
  - Layers: 12 - Token (hidden) size: 768 - Self-Attention heads: 12 - Parameters: 117M
- Pretraining
  - The BooksCorpus (800M words)
- GPT family

| Model                                               | Year      | Parameters |
|-----------------------------------------------------|-----------|------------|
| GPT-1                                               | 2018      | 117M       |
| GPT-2 (largest                                      | 2019      | 1.5B       |
| GPT-3 (largest)                                     | 2020      | 175B       |
| GPT-3.5 – used in first ChatGPT release (Nov. 2022) | 2022      | ~175B      |
| GPT-4                                               | 2023      | 100B–1T    |
| GPT-5-class (ChatGPT today)                         | 2025–2026 | >1T        |



# BERT - Bidirectional Encoder Representations from Transformers

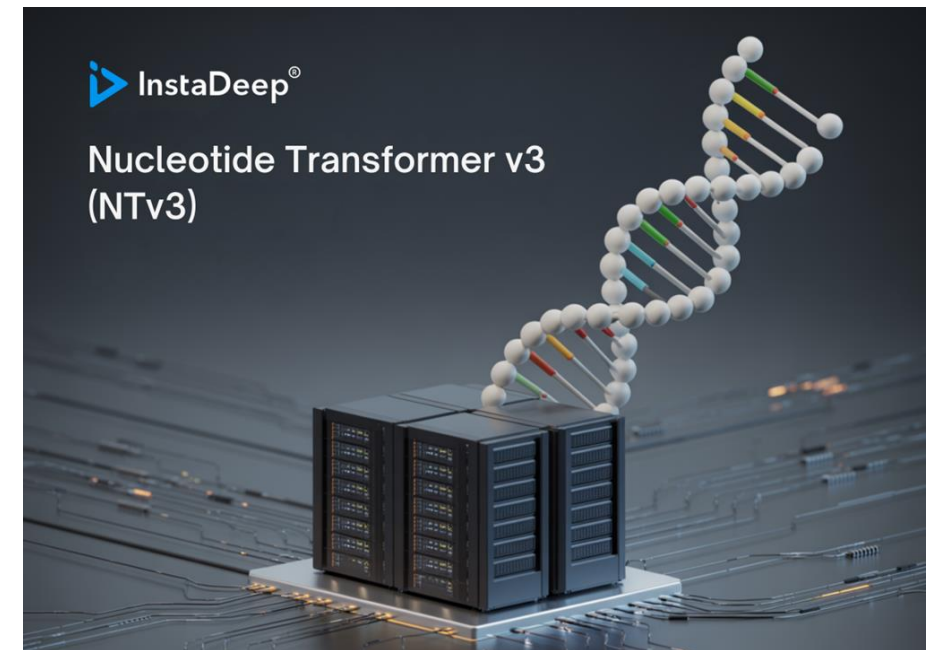
‘Universal language understanding model’ (2018) [13] by Google AI

- Architecture
  - Encoder-only transformer
  - Bidirectional: learn information from a text from the left and right side
  - BERT<sub>BASE</sub>: Layers: 12 - Token (hidden) size: 768 - Self-Attention heads: 12 - Parameters: 110M
  - BERT<sub>LARGE</sub>: Layers: 24 - Token (hidden) size: 1024 - Self-Attention heads: 16 - Parameters: 340M
- Pretraining
  - The BooksCorpus (800M words) & English Wikipedia (2,500M words)
  - 4 days: BERT<sub>BASE</sub> on 16 TPUs                      BERT<sub>LARGE</sub>: on 64 TPUs (<\$1M capital cost)
- Fine-tuned on 11 different NLP tasks
- Many BERT variants including RoBERTa, ALBERT, DistilBERT, ELECTRA
  - **DNABERT-2 (2024)** [14] based on BERT<sub>BASE</sub> (~117M parameters) – *TinyDNABERT to be used in workshop* but **BPE tokenisation** and **trained on multi-species genomes** (810+)
  - Most LLMs for genomics rely on a BERT-like architecture, including Nucleotide Transformer (2025)

# Nucleotide Transformer

Large-scale, high-capacity genomics foundation model (2025) [15]  
by InstaDeep + NVIDIA + academia

- Architecture
  - Encoder-only transformer & bidirectional (similar to BERT)
  - From 6 to 36 layers, from 50M to 2.5B parameters
  - Tokenization: 6-mer non-overlapping tokens
- Pretraining
  - Multi-species genomes (850+)
  - Training time: not reported, but used Cambridge-1 Supercomputer (~\$100M, ~8 PFLOPS)



*to be used in guided hands-on session*

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